



JURONG JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2018

CANDIDATE
NAME

CLASS

18S

EXAM INDEX

CHEMISTRY

Higher 2

Paper 3 Free Response

9729/03

11 September 2018

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams, graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A *Data Booklet* is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 (a) Peroxides refer to a class of compounds with an oxygen-oxygen single bond.

Hydrogen peroxide, H_2O_2 , is the simplest peroxide. It is commonly used in laboratories for both its oxidising and reducing properties.

Sodium peroxide, Na_2O_2 , is prepared by burning sodium in air. This reaction also produces sodium oxide, Na_2O .

Compound	Melting point / $^{\circ}\text{C}$
hydrogen peroxide, H_2O_2	-11
sodium peroxide, Na_2O_2	675
sodium oxide, Na_2O	920

- (i) Draw the 'dot-and-cross' diagrams for

- hydrogen peroxide, H_2O_2
- sodium peroxide, Na_2O_2 .

[2]

- (ii) Explain the difference in the melting points between

- sodium peroxide (Na_2O_2) and hydrogen peroxide (H_2O_2)
- sodium peroxide (Na_2O_2) and sodium oxide (Na_2O).

[2]

- (b) The peroxides of the Group 2 elements, MO_2 , decompose on heating to produce a single gas and the solid oxide, MO , only.

- (i) Write an equation for the thermal decomposition of strontium peroxide, SrO_2 .

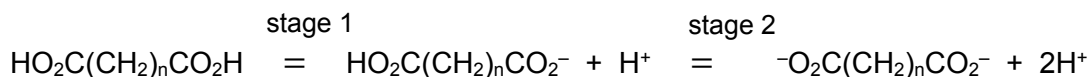
[1]

- (ii) Suggest how the temperature at which thermal decomposition of MO_2 occurs varies down Group 2.

Explain your answer.

[3]

1 (c) Dicarboxylic acids dissociate in stages.

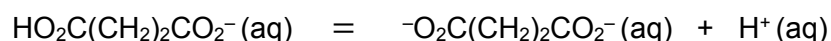


The pK_a values for stage 1 and stage 2 for some dicarboxylic acids are listed below.

n in $\text{HO}_2\text{C}(\text{CH}_2)_n\text{CO}_2\text{H}$	$pK_a(1)$ for stage 1	$pK_a(2)$ for stage 2
1	2.83	5.69
2	4.16	5.61
3	4.31	5.41

For comparison, the pK_a value of ethanoic acid, $\text{CH}_3\text{CO}_2\text{H}$, is 4.76.

- (i) With reference to the table given, suggest why the $pK_a(1)$ values
- are all smaller than the pK_a of ethanoic acid,
 - become larger as n increases. [3]
- (ii) Suggest why all the $pK_a(2)$ values in the table above are larger than the pK_a of ethanoic acid. [1]
- (iii) Calculate the pH of a 0.10 mol dm^{-3} solution of $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$. Ignore the effect of $pK_a(2)$ on the pH. [1]
- (iv) Sketch the pH-volume added curve you would expect to obtain when 20 cm^3 of 0.10 mol dm^{-3} NaOH is added to 10 cm^3 of 0.10 mol dm^{-3} $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$.
Mark clearly the initial pH and the point(s) of maximum buffering capacity. Indicate the volume at equivalence point(s). [3]
- (v) The monosodium salts of edible dicarboxylic acids can act as buffers.
Write two equations to show how monosodium butanedioate, $\text{HO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{Na}$, acts as a buffer. [2]
- (vi) A solution containing both $\text{HO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{Na}$ and $\text{NaO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{Na}$ forms a buffer solution. The following equilibrium is present in the solution.



By choosing the correct pK_a value given in the table, calculate the pH of a buffer solution made by mixing 100 cm^3 of 0.5 mol dm^{-3} $\text{HO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{Na}$ and 50 cm^3 of 0.3 mol dm^{-3} $\text{NaO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{Na}$. [2]

[Total: 20]

2 Sulfur is the second member of Group 16 in the Periodic Table.

(a) Explain why sulfur has a lower first ionisation energy than phosphorus. [1]

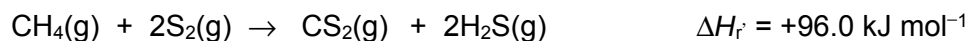
(b) Natural gas, consisting primarily of methane, is available in large amounts and is used as a fuel. However, direct conversion of alkanes such as methane into useful products is challenging owing to their unreactivity.

(i) Give a reason why alkanes are unreactive. [1]

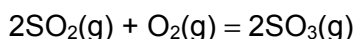
In 2013, scientists discovered that they could selectively convert methane to alkenes using gaseous sulfur, S_2 , over a suitable catalyst. This conversion takes place at over 1000 K, where sulfur exists as $S_2(g)$, as shown in **reaction 1**.

(ii) **reaction 1** $2CH_4(g) + S_2(g) \rightarrow C_2H_4(g) + 2H_2S(g)$

Calculate the enthalpy change of **reaction 1** by drawing an energy cycle using the following information.



(c) The key stage in the manufacture of sulfuric acid is the reaction between sulfur dioxide and oxygen over a catalyst.



(i) When an equimolar mixture of SO_2 and O_2 is passed over a catalyst at $T^\circ\text{C}$ at an initial total pressure of 200 kPa, the percentage conversion of $SO_2(g)$ is 98%.

Calculate the equilibrium partial pressure of each of the three gases and hence, the value of K_p at temperature $T^\circ\text{C}$. [3]

(ii) The table below shows values of K_p for this equilibrium at different temperatures.

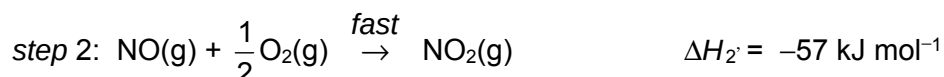
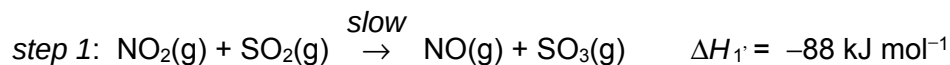
temperature / $^\circ\text{C}$	K_p / kPa^{-1}
25	4.0×10^{22}
200	2.5×10^8
800	1.3×10^{-3}

Using relevant data from the given table, explain in terms of the position of equilibrium,

- the sign of ΔG° at 25°C and
- the sign of ΔH for the forward reaction. [4]

- 2 (d) (i) Which of the two gases, SO_2 and O_2 , is less ideal? Explain your answer. [1]
- (ii) Sketch a graph of volume (V) against temperature (T/K) for a given mass of an ideal gas at constant pressure. [1]
- (e) In the atmosphere, the oxidation of SO_2 to SO_3 can be catalysed by NO_2 .

The mechanism for this catalysed oxidation of SO_2 to SO_3 occurs in two steps.



Use this information to construct a fully-labelled reaction pathway diagram for the catalysed oxidation of SO_2 to SO_3 . Indicate on your diagram, the 'reactants', 'products', ΔH_1 , ΔH_2 and the activation energy, E_a , of the reaction. [3]

- (f) Oxides of nitrogen are also used in other applications.

One such oxide is N_2O_5 , which exists as $[\text{NO}_2^+][\text{NO}_3^-]$ in solution.

When benzene is added to a solution of N_2O_5 in CCl_4 , an excellent yield of nitrobenzene is obtained in the absence of any added catalyst.

- (i) Write a balanced equation for the reaction of benzene and N_2O_5 . [1]
- (ii) Suggest a mechanism for this reaction, showing all charges and using curly arrows to show the movement of electron pairs. [2]
- (iii) Compound **H** has the molecular formula, C_9H_{12} . Mononitration of **H** produces only one organic product. Suggest the structure of **H**. [1]

[Total: 20]

- 3 (a) Copper has the highest electrical conductivity rating among the non-precious metals. The high conductivity of copper is a consequence of its electronic configuration.

(i) State the electronic configuration of Cu atom. [1]

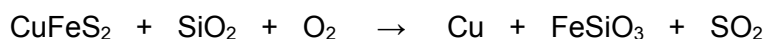
(ii) Explain why copper is regarded as a *transition element*. [1]

(iii) Suggest why copper is **not** usually used for overhead electrical cables despite its superior conductivity. [1]

- (b) Electrical wiring is the most important market for the copper industry. To be used in wiring, copper must be at least 99.9% pure. Chalcopyrite, CuFeS_2 , is the most common copper ore being used to obtain pure copper in a two-step process.

The first step occurs in a furnace where chalcopyrite is heated strongly with silica, SiO_2 , and air. The furnace reduces the copper(II) in chalcopyrite to copper.

The reaction occurring in the blast furnace can be represented by the following **unbalanced** equation.



The copper obtained is nowhere near 99.9% pure as it contains nickel and silver as minor impurities to form an alloy.

The second step occurs in an electrolytic cell where the alloy undergoes electrolysis to produce copper at or above 99.9% purity.

(i) Identify the element that undergoes oxidation in the furnace and state the initial and final oxidation numbers of this element. [1]

(ii) In the furnace, the element O is reduced alongside with the element Cu. Using oxidation numbers or otherwise, write a **balanced** equation for the reaction occurring in the furnace. [1]

(iii) Draw a fully labelled diagram of the electrolytic cell where copper purification occurs. [2]

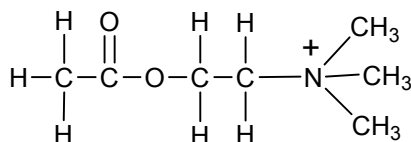
(iv) Explain, by quoting relevant E° values, what happens to the nickel and silver impurities during this purification procedure. [4]

- (c) When dilute aqueous ammonia was added to copper(II) sulfate solution, a pale blue precipitate was first observed which dissolved to give a dark blue solution when aqueous ammonia was added in excess. In these reactions, the ammonia can act as a *Bronsted-Lowry base* and as a *Lewis base*.

Illustrate the meaning of *Bronsted-Lowry base* and *Lewis base* using the reactions of aqueous copper(II) ions with ammonia. Write equations to account for the observations noted in these reactions. [4]

- 3 (d) In 2013, scientists suggested that too much copper in our diet may contribute to Alzheimer's disease, a neurological disorder in which death of brain cells causes memory loss and cognitive decline. In patients with Alzheimer's disease, the concentration of acetylcholine, a neurotransmitter essential for processing memory and learning, is found to be lower.

The structure of acetylcholine is shown below.

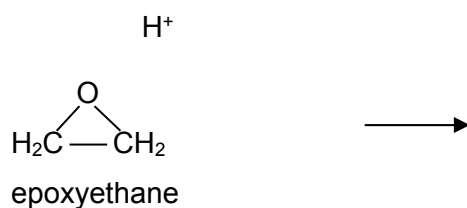


When acetylcholine undergoes hydrolysis in the presence of an enzyme, two products are formed; one of which is a charged species called choline.

- (i) Draw the structures of these **two** products. [2]

- (ii)** As a precursor to the synthesis of acetylcholine, choline is an essential nutrient in our diet. Choline can be produced in a sealed reaction vessel by the reaction of anhydrous trimethylamine, $\text{N}(\text{CH}_3)_3$, with epoxyethane and HCl under moderate pressure.

Copy and complete the diagram below to suggest a mechanism to show how choline is formed. Show all charges and relevant lone pairs and show the movement of electron pairs by using curly arrows.



[3]

[Total: 20]

Section B

Answer **one** question from this section.

- 4 (a) Thionyl chloride, SOCl_2 , is a colourless liquid that is primarily used in the chlorination of organic compounds.

(i) Draw the shape of SOCl_2 . [1]

(ii) Explain which molecule, SOCl_2 or COCl_2 , is expected to have a larger bond angle. [1]

- (b) When SOCl_2 is reacted with a carboxylic acid to produce an acyl chloride, two acidic gases are formed.



A 1.00 g sample of a carboxylic acid RCO_2H was treated in this way, and the gases were absorbed in 60.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ NaOH(aq) . The excess NaOH was titrated with $0.500 \text{ mol dm}^{-3}$ HCl(aq) . It required 10.8 cm^3 of the HCl(aq) solution to reach the end-point.

- (i) Write equations for the complete reactions between

- NaOH and HCl ,
- NaOH and SO_2 . [2]

(ii) Calculate the total number of moles of NaOH that reacted with the acidic gases, SO_2 and HCl . [1]

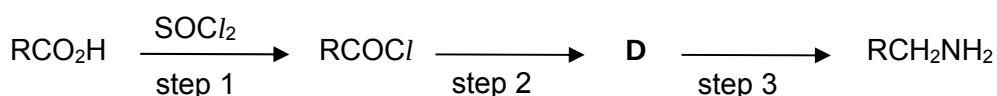
(iii) Calculate the number of moles of RCO_2H that produced the SO_2 and HCl . [1]

(iv) Hence calculate the M_r of the carboxylic acid, RCO_2H . [1]

(v) The R group contains carbon and hydrogen only.

Suggest the structure of RCO_2H . [1]

- (c) Using SOCl_2 in the initial step, carboxylic acids can be converted into primary amines by the following sequence of reactions.



(i) Identify the structure of D. [1]

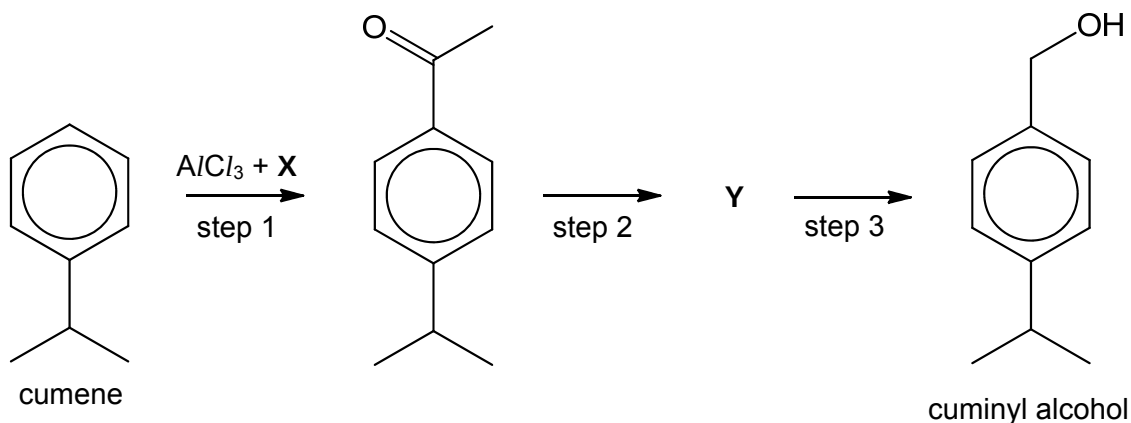
(ii) Suggest the reagents needed for step 2 and step 3. [2]

(iii) State the *types of reaction* for step 2 and step 3. [1]

- 4 (d) Angelic acid, $C_5H_8O_2$, is a natural product isolated from the roots of the angelica plant.
- Angelic acid reacts with $H_2 + Ni$ to form **T**, $C_5H_{10}O_2$.
 - Both angelic acid and **T** exhibit stereoisomerism.
 - On treatment with hot acidic $KMnO_4$, angelic acid produces two acidic compounds, **U** and **V**.
 - Only **U** gives yellow precipitate with alkaline aqueous iodine.
- (i) Suggest structures for **T**, **U**, **V** and angelic acid. Explain the reactions. [6]
- (ii) State the types of stereoisomerism shown by angelic acid and compound **T**. [2]

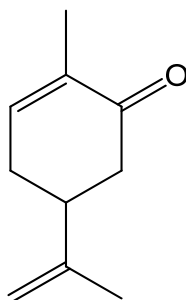
[Total: 20]

- 5 (a) (i) Describe the reaction of SiCl_4 with water and include any observations you can make. Write an equation for the reaction that occurs. [2]
- (ii) Carbon is in the same group as silicon. However, CCl_4 does not react with water. Explain why. [1]
- (b) Chlorine-containing organic compounds can also differ in their reactivity towards hydrolysis reactions.
- Compounds **A** and **B** are isomers with the molecular formula, $\text{C}_7\text{H}_7\text{Cl}$. Compound **A** reacts with aqueous NaOH on heating but compound **B** does not.
- Draw the structures of compounds **A** and **B** and explain the difference in their reactivity towards aqueous NaOH . [2]
- (c) Heating tin with hydrochloric acid produces hydrogen gas. Careful removal of water from the resulting product produces white solid SnCl_2 .
- In contrast, passing chlorine gas over heated tin produces colourless liquid SnCl_4 as the only product.
- Using relevant E° from the *Data Booklet*, explain the above observations. [2]
- (d) Cumin is a spice used to flavour food. One of the key organic compounds responsible for the smell of cumin is cumyl alcohol. It can be synthesised from cumene as shown below. Aluminium chloride is used as a catalyst with reagent **X** in step 1.



- (i) Suggest the identity of reagent **X**. [1]
- (ii) Suggest reagents and conditions for steps 2 and 3 and draw the structure of the intermediate compound **Y**. [3]

- 5 (e) Carvone, an isomer of cuminyl alcohol, is most responsible for the flavour of caraway, dill and spearmint.



carvone

- (i) When carvone is treated with NaBH_4 , compound **M**, $\text{C}_{10}\text{H}_{16}\text{O}$, is produced. Draw the structure of **M** and write an equation to represent this reaction. [2]

- (ii) Compound **N** is isomeric with compound **M**.

On treatment with hot concentrated KMnO_4 , **N** gives two compounds, **P**, $\text{C}_3\text{H}_6\text{O}$, and **Q**, $\text{C}_4\text{H}_4\text{O}_5$. **P** and **Q** are formed in the mole ratio of 2 : 1.

All three compounds, **N**, **P** and **Q** give an orange precipitate with 2,4-dinitrophenylhydrazine but only **N** forms a silver mirror with Tollens' reagent. One mole of **Q** reacts with two moles of aqueous sodium hydrogencarbonate.

Suggest structures for **N**, **P** and **Q** and explain the observations described above. [7]

[Total: 20]

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